

Global-Sign Correlation Breaks BiT-128

Practical Equivalent-Key Recovery and Chosen-Message Forgery

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Provenance and review status. The attack, derivation, implementation, experiments, and initial manuscript were produced independently by OpenAI Codex using Daybreak Blue at max reasoning effort during an AI-run audit of NGCC signature candidates. At the date above, no human cryptanalyst has yet independently verified the argument, implementation, or measurements. This report should therefore be treated as a preliminary disclosure for human review and reproduction.

Abstract

BiT is a Fiat–Shamir-with-aborts lattice signature submitted to the Next-generation Commercial Cryptographic Algorithms Program. Its response is formed as $z = y + rsc$, where a single hidden sign $r \in \{-1, +1\}$ is shared by all coefficients. BiT then applies a one-dimensional rejection rule to every coefficient separately. We show that this rule produces the advertised one-dimensional marginals but does not remove the joint dependence created by the shared sign. The defect is especially damaging because the response to the public constant component of the secret is included in every signature. For BiT-128, its 30 challenge positions identify the hidden sign with about 96.2% hard-decision accuracy; the posterior mean retains about 88.9% of the information of a revealed sign.

We turn the leakage into a streaming weighted regression over $\mathbb{Z}[x]/(x^{256} + 1)$. Negacyclic Fourier diagonalisation recovers the three ternary secret polynomials from ordinary signatures. A small search over the least certain rounded coefficients is checked using the compressed public key. Once those polynomials are known, public-key compression permits construction of a short error and low part that form an equivalent signing key.

An end-to-end experiment against the submitted BiT-128 reference implementation used 200,000 signatures. The direct estimator left five of 768 secret coefficients incorrect. Public-key correction found the valid assignment after 1,596 subset tests. The recovered secret was completed to a different valid signing key, and the unmodified submitted verifier accepted a 1,504-byte signature on a fresh message. The entire single-threaded run, including reference signing, used 139 CPU-seconds on a 3.2 GHz Intel Core i5-6500 and negligible attack memory. The attack is information-theoretic, does not exploit the hash function, and remains valid in the random-oracle model. The fixed 200,000-query procedure forged on three of four independently generated keys; the fourth forged after extending the transcript to 250,000 signatures.

1 Introduction

Fiat–Shamir-with-aborts signatures use rejection sampling to make a response statistically independent of the signing key. That independence is a joint distribution requirement: an adversary receives a complete vector, not a collection of isolated scalar samples. Replacing a

vector rejection ratio by the product of ratios for its scalar marginals is generally invalid when a latent random variable is shared by all coordinates.

BiT (“Bimodal Triangular distribution based lattice signatures”) is an NGCC digital-signature candidate built around exactly such a latent variable [1]. It samples one bit b , writes $r = (-1)^b$, and forms

$$z = y + rsc.$$

The mask coefficients are independent triangular samples, but the sign r is global. BiT’s rejection procedure derives the correct scalar mixture for each coordinate and accepts each coordinate with the corresponding scalar ratio. This proves a marginal statement. It does not transform the vector mixture into a secret-independent product distribution.

The distinction is directly exploitable. BiT’s extended secret is $s = (1, s_0, e)$. Its first response polynomial therefore contains $z_{1,0} = y_{1,0} + rc$. At each nonzero challenge position the offset is the known value r , so these public response coefficients act as repeated noisy observations of the global sign. Once a posterior for r is available, the remaining public response polynomials $z_{1,1} = y_{1,1} + r(s_0c)$ become noisy linear observations of s_0 .

Our contributions are as follows.

- We identify the joint-distribution error in BiT’s coordinatewise bimodal rejection sampler and give an explicit expression for the residual dependence.
- We derive a sign distinguisher from $z_{1,0}$. For the BiT-128 parameters it has about 96.2% hard-decision accuracy from one signature.
- We give a streaming key-recovery algorithm based on posterior-weighted regression. A negacyclic Fourier transform reduces recovery to 768 scalar estimates and requires only $O(N)$ stored state per secret polynomial.
- We show how to validate and correct uncertain coefficients using the compressed public key, and how to complete recovered s_0 to an equivalent signing key without recovering the original error e .
- We demonstrate an accepted fresh-message forgery against the submitted BiT-128 implementation using 200,000 signatures and about 2.3 minutes of single-threaded CPU time on an ordinary 2015 desktop processor.

This is a full key-recovery and forgery attack on the claimed 128-bit instance. It is not a distinguisher, a side channel, or a reduction in an asymptotic security estimate. The attack needs no control over messages beyond obtaining distinct valid signatures; it can therefore be mounted passively when enough known-message signatures are available.

2 The relevant part of BiT

We describe only the components needed for the attack. The notation and parameter names follow the submission specification [1]. Let

$$\mathcal{R}_q = \mathbb{Z}_q[x]/(x^N + 1), \quad N = 256, \quad q = 26881$$

for BiT-128. Multiplication by a challenge polynomial c is a negacyclic linear map, which we denote by C when using coefficient vectors.

2.1 Keys and public-key compression

The secret consists of $s_0 \in S_1^m$ and $e \in S_1^n$, where every coefficient lies in $\{-1, 0, 1\}$ and $(m, n) = (3, 3)$. With public matrix $A_0 \in \mathcal{R}_q^{n \times m}$, key generation computes

$$u = A_0 s_0 + e, \quad (b_1, b_0) = (\text{HighBits}(u, \gamma_b), \text{LowBits}(u, \gamma_b)), \quad \gamma_b = 16.$$

The public key contains the seed defining A_0 and b_1 ; the signing key also contains s_0, e, b_0 . Defining

$$A = [\hat{q}j_n - \gamma_b b_1, A_0, I_n], \quad s = (1, s_0^T, e^T)^T,$$

where $\hat{q} = 2^{-1} \bmod q$, gives $As = \hat{q}j_n + b_0$.

2.2 Signing and the shared sign

For a message, the signer samples independent triangular masks

$$y = (y_{1,0}, y_{1,1}, y_2) \leftarrow \mathsf{T}_{\gamma_{1,0}} \times \mathsf{T}_{\gamma_{1,1}}^m \times \mathsf{T}_{\gamma_{1,2}}^n.$$

It hashes the high part of Ay and the message representative to obtain a challenge c of Hamming weight τ . The submitted BiT challenge has coefficients in $\{0, 1\}$, with exactly $\tau = 30$ ones for BiT-128. One bit b is sampled and the global sign $r = (-1)^b$ is applied to all components:

$$\begin{aligned} z_{1,0} &= y_{1,0} + rc, \\ z_{1,1} &= y_{1,1} + r(s_0 c), \\ z_2 &= y_2 + r(ec). \end{aligned}$$

The signer applies coefficientwise rejection, performs further high-bits and hint checks, and returns $(z_1, h, \tilde{c})$, where $z_1 = (z_{1,0}, z_{1,1})$. The z_2 component is compressed into the hint h .

Table 1 lists the parameters that control the attack.

Table 1: Relevant submitted BiT-128 parameters.

Parameter	Value	Role
N, q	256, 26881	ring degree and modulus
(n, m)	(3, 3)	module dimensions
η	1	ternary secret bound
τ	30	challenge weight
$\gamma_{1,0}$	8	mask for the known constant component
$\gamma_{1,1}$	1024	mask for $s_0 c$
$\gamma_{1,2}$	2048	mask for ec
γ_b	16	public-key compression interval
Public key	1048 bytes	submitted encoding
Signature	1504 bytes	submitted encoding

2.3 Scalar triangular rejection

Ignoring a common normalising factor, the triangular mass function is

$$f_\gamma(x) = \max\{\gamma - |x|, 0\}.$$

For a scalar secret offset v , the source marginal after hiding the sign is

$$F_{\gamma,v}(x) = \frac{f_{\gamma}(x-v) + f_{\gamma}(x+v)}{2}.$$

BiT defines a target $G_{\gamma,\beta}(x)$ using a public bound β and accepts a scalar response with probability proportional to

$$a_v(x) = \frac{G_{\gamma,\beta}(x)}{F_{\gamma,v}(x)}.$$

The procedure is then applied separately to every coefficient. At the scalar level this is correct: averaging over r transforms $F_{\gamma,v}$ into the chosen target. The next section shows why applying these ratios independently does not hide a sign shared by the entire vector.

3 The joint-distribution flaw

Consider d response coordinates with independent masks and secret offsets $v = (v_1, \dots, v_d)$, but a single sign r . Before rejection, the joint source is

$$P_v(z) = \frac{1}{2} \prod_{i=1}^d f_i(z_i - v_i) + \frac{1}{2} \prod_{i=1}^d f_i(z_i + v_i). \quad (1)$$

It is not the product of scalar mixtures $\prod_i F_i(z_i)$. Nevertheless, BiT applies the acceptance product

$$a_v(z) = \prod_{i=1}^d \frac{G_i(z_i)}{F_i(z_i)}. \quad (2)$$

For a fixed sign, define the accepted one-coordinate law

$$q_{i,r}(x) = \frac{f_i(x - rv_i)G_i(x)/F_i(x)}{\mathcal{Z}_i}, \quad \mathcal{Z}_i = \sum_x f_i(x - v_i)G_i(x)/F_i(x). \quad (3)$$

Symmetry gives the same normaliser for $r = +1$ and $r = -1$, and $q_{i,-}(x) = q_{i,+}(-x)$.

Proposition 1 (Marginals are hidden but the joint law is not). *After BiT's coordinatewise rejection, the response law is*

$$Q_v(z) = \frac{1}{2} \prod_{i=1}^d q_{i,+}(z_i) + \frac{1}{2} \prod_{i=1}^d q_{i,-}(z_i). \quad (4)$$

Every scalar marginal equals the advertised target $G_i/\mathcal{Z}_i$, but, except in degenerate cases, the joint distribution depends on v .

Proof. Conditioned on r , independent masks and independent rejection coins factor, which gives the two products in (4). Averaging one coordinate over the two signs gives

$$\frac{q_{i,+}(x) + q_{i,-}(x)}{2} = \frac{G_i(x)}{\mathcal{Z}_i}.$$

The mixture of two products in (4) is generally not the product of these averages. For example, let X_i denote the i th accepted response coordinate and put $\mu_i(v_i) = \mathbb{E}[X_i \mid r = +1]$. Conditional independence and symmetry give, for distinct coordinates,

$$\mathbb{E}[X_i X_j] = \mu_i(v_i) \mu_j(v_j), \quad \mathbb{E}[X_i] = \mathbb{E}[X_j] = 0.$$

Thus the shared sign leaves a secret-dependent cross-coordinate covariance even though both scalar marginals have the intended distribution. $\square$

A correct vector rejection sampler would have to use the true source (1) in a joint ratio. Multiplying ratios derived from the scalar mixtures silently replaces a shared latent sign by independent latent signs, which is a different distribution.

4 Recovering the hidden sign from the known component

BiT makes the joint leak unusually accessible because the first component of the extended secret is the public constant 1. At each challenge position,

$$X = z_{1,0,i} = Y + r,$$

with mask width $\gamma_{1,0} = 8$. From (3), the scalar likelihood ratio is

$$\frac{q_+(x)}{q_-(x)} = \frac{f_8(x-1)}{f_8(x+1)}. \quad (5)$$

The target mass and acceptance probability cancel. For the 15 possible values $x = -7, \dots, 7$, the conditional numerator masses are

$$\begin{aligned} 63q_+(x) &= (0, 1, 2, 3, 4, 5, 6, 7, 8, 7, 6, 5, 4, 3, 2), \\ 63q_-(x) &= (2, 3, 4, 5, 6, 7, 8, 7, 6, 5, 4, 3, 2, 1, 0). \end{aligned} \quad (6)$$

In particular, $x = 7$ determines $r = +1$ and $x = -7$ determines $r = -1$.

A signature contains $\tau = 30$ such samples sharing the same r . Let S_t be the support of its challenge. The log-likelihood ratio and posterior mean are

$$\Lambda_t = \sum_{i \in S_t} \log \frac{f_8(z_{1,0,i} - 1)}{f_8(z_{1,0,i} + 1)}, \quad (7)$$

$$w_t = \mathbb{E}[r_t \mid z_{1,0}, c_t] = \tanh(\Lambda_t/2). \quad (8)$$

Numerical convolution of the finite law (6) gives a Bayes hard-decision success probability of approximately 0.962. It also gives

$$\mathbb{E}[w_t^2] \approx 0.8894. \quad (9)$$

The latter is the useful quantity for soft-decision recovery. A value of one would correspond to a fully revealed sign, while zero would convey no sign information. Thus one BiT-128 signature retains almost 89 percent of the linear-regression information available if b were explicitly published.

The additional BiT rejection and hint conditions do not repair this defect. In our final-signature measurements the average w_t^2 was 0.8896, matching (9) to four significant figures. In a separate 20,000-signature run instrumented only to record the sign after each signature had already been accepted, the hard decision $\text{sign}(w_t)$ was correct in 96.265% of signatures and $Q^{-1} \sum_t w_t r_t = 0.889743$. The attack itself does not use this instrumentation.

5 Streaming recovery of the ternary secret

Let $s_j \in \{-1, 0, 1\}^N$ be one of the three polynomials of s_0 , and let C_t be negacyclic multiplication by the public challenge c_t . Conditional on r_t , each accepted response coordinate has mean

$$m(v) = \frac{\sum_x x f_{1024}(x-v) G_{1024,30}(x) / F_{1024,v}(x)}{\sum_x f_{1024}(x-v) G_{1024,30}(x) / F_{1024,v}(x)}. \quad (10)$$

This is odd in v . On the complete range $|v| \leq 30$, it is close to the identity: $m(1) = 0.99987$ and $m(30) = 29.1461$. The convolution values arising from ternary secrets and weight-30 challenges are concentrated much nearer zero, where the approximation $m(v) \simeq v$ is tighter.

Using the posterior (8), the public response model is therefore

$$z_{t,j} \simeq w_t C_t s_j + \varepsilon_{t,j}. \quad (11)$$

The noise has standard deviation about 418 per coordinate, but each signature supplies $N = 256$ equations for every secret polynomial. Ordinary weighted least squares gives

$$\tilde{s}_j = \left(\sum_t w_t^2 C_t^* C_t \right)^{-1} \left(\sum_t w_t C_t^* z_{t,j} \right). \quad (12)$$

The operators C_t are negacyclic circulants and are diagonal at the roots of $X^N + 1$. If hats denote this negacyclic Fourier transform, then every frequency can be estimated independently:

$$\widehat{s}_{j,k} = \frac{\sum_t w_t \overline{\widehat{c}_{t,k}} \widehat{z}_{t,j,k}}{\sum_t w_t^2 |\widehat{c}_{t,k}|^2}. \quad (13)$$

The attack streams signatures through these numerator and denominator sums; it need not store the signatures. After an inverse transform, each real coefficient is rounded to $\{-1, 0, 1\}$.

For Q signatures, the arithmetic cost is $O(Q(m+1)N \log N)$ and the stored attack state is $O(mN)$. These costs are small beside acquiring signatures. The attack uses the challenge and response already present in the public signature and never uses a signing-time measurement.

6 Public correction and equivalent-key completion

At practical query counts, a few estimates may lie on the wrong side of a ternary rounding boundary. The public key both identifies those errors and allows completion to a signing key.

6.1 Low-confidence subset search

For an estimate a , define its distance to the nearest relevant decision boundary by

$$d(a) = ||a| - 1/2|.$$

Sort all 768 estimates by $d(a)$. For a coefficient rounded to ± 1 , the alternate value is zero. For a coefficient rounded to zero, the alternate is the adjacent sign indicated by a . We enumerate subsets of the 24 least certain alternates in Gray-code order.

Each assignment determines $A_0 s_0$. Before performing a full check, three coefficients are tested against the public condition

$$\exists e' \in \{-1, 0, 1\} : \text{HighBits}((A_0 s_0)_i + e', 16) = (b_1)_i. \quad (14)$$

For a wrong assignment, one such equality holds with probability roughly the width of one compression interval divided by q ; three independent checks therefore discard essentially every wrong subset. Gray-code updates require only one precomputed column contribution per subset.

6.2 Completing an equivalent signing key

Suppose s_0 is correct. For each coefficient, choose any $e'_i \in \{-1, 0, 1\}$ satisfying (14) and set

$$b'_0 = \text{LowBits}(A_0 s_0 + e', 16). \quad (15)$$

Such a choice always exists because the original error is one witness. It is usually possible to choose $e'_i = 0$; only coefficients near compression boundaries require ± 1 . Then

$$A_0 s_0 + e' = 16b_1 + b'_0,$$

so (s_0, e', b'_0) satisfies exactly the key relation used by signing and verification. The signing-randomness seed can be chosen afresh, and the public key hash is recomputed from the public key. This yields an equivalent secret key, although e' need not equal the key generator’s original e .

This completion is also a public validation test for recovery. An incorrect ternary vector makes $A_0 s_0$ pseudorandomly inconsistent with hundreds of compressed public coefficients; false acceptance is negligible.

7 End-to-end experiment

7.1 Method

We evaluated the attack against the submitted BiT-128 reference algorithm with its default SM3-based hashing and exact submitted encodings. One key pair was generated, and the signing oracle was queried on distinct 24-byte messages containing a counter. Recovery consumed only the public key and the resulting signatures. The original secret was retained outside the attack and used only after each public validation step to count diagnostic coefficient errors.

The implementation used double-precision radix-2 transforms at the 256 roots of $X^{256} + 1$. It accumulated (13), rounded the inverse transform, and applied the 24-position public correction described above. A fresh equivalent key was packed using (15). The forgery message was distinct from every oracle query, and the final verdict was obtained from the submitted verifier without modification.

The experiment ran single-threaded on an Intel Core i5-6500 at 3.2 GHz with 16 GiB RAM, using GCC 13.3.0 with optimisation enabled. CPU time was measured over key recovery and the reference signing calls. Attack state consists of a few arrays of 256 complex doubles and is far below one megabyte; the reference signer and temporary objects also fit easily in ordinary laptop memory.

7.2 Results

Table 2 gives the complete 200,000-signature run. The direct estimator’s coefficient RMSE was 0.186328. Only five ternary coefficients rounded incorrectly. All five occurred among the 24 positions selected solely by public confidence scores. The Gray-code enumeration found the publicly valid assignment after 1,596 subset tests, far short of its 2^{24} cap.

Table 2: End-to-end BiT-128 attack result.

Quantity	Result
Signature queries	200,000
Posterior mean $Q^{-1} \sum_t w_t^2$	0.889585
Direct-estimate RMSE	0.186328
Wrong coefficients before correction	5 of 768
Low-confidence positions searched	24
Subset tests before public match	1,596
Wrong coefficients after correction	0 of 768
Nonzero coefficients in completed e'	31 of 768
CPU time including signing	139.3 seconds
Forged signature size	1,504 bytes
Fresh-message verifier result	accepted

The completed key deliberately preferred $e'_i = 0$ whenever possible and had only 31 nonzero error coefficients. It was therefore plainly different from a normally sampled error polynomial, yet it met the public relation and signed correctly. The verifier accepted the resulting signature on the unqueried message. This demonstrates the complete chain

$$\text{signatures} \longrightarrow s_0 \longrightarrow (s_0, e', b'_0) \longrightarrow \text{fresh forgery}.$$

The measured RMSE followed the expected $Q^{-1/2}$ trend: it was 0.3763 at 50,000 queries, 0.2743 at 100,000, 0.2166 at 150,000, and 0.1863 at 200,000. Public correction is therefore doing bounded finishing work rather than compensating for a nonconvergent estimator.

7.3 Additional-key checks

We repeated the fixed 200,000-query procedure on three further keys, without changing the 24-position correction list. Table 3 reports all four completed key trials, including the main experiment above. Three recovered an equivalent key and produced an accepted fresh-message forgery. On trial D, at least one of the five remaining errors lay outside the 24 selected positions, so the fixed correction search exhausted its cap. The failure is reported rather than discarded; an attacker can respond by collecting more signatures or enlarging the public correction list.

Table 3: Fixed-parameter checks on four independently generated BiT-128 keys.

Trial	RMSE	Direct errors	Subsets tested	Result
A	0.186328	5	1,596	accepted forgery
B	0.185175	8	41,055	accepted forgery
C	0.185637	4	16,131	accepted forgery
D	0.194541	5	16,777,216	correction cap reached

This small sample is a robustness check, not a precise estimate of attack success probability. It shows both that the main result is reproducible on other keys and that 200,000 queries with a 24-position list is not a deterministic success threshold. Continuing trial D to 250,000 signatures reduced its direct estimate to one error with RMSE 0.173932; the same 24-position correction then found the exact key after 256 subset tests, and the fresh-message forgery was accepted. Thus all four tested keys were broken with at most 250,000 signatures.

7.4 Why this is a mathematical attack

No timing, cache trace, fault, malformed input, or implementation-specific undefined behaviour is used. The only inputs to recovery are ordinary public keys and valid signatures. Hash outputs are treated as independent random challenges; the attack does not predict, collide, or invert SM3. The leak is present in the specified probability distribution, so replacing the hash or making the implementation constant-time does not address it.

8 Other parameter sets

The same joint-distribution defect is present whenever one global sign is combined with coordinatewise scalar rejection. BiT-256 and BiT-512 also publish the response to the known constant component, so the sign-posterior step applies directly. Their parameters use larger masks for s_0c , which raises the number of signatures needed by the simple regression in this paper. We therefore claim a measured full break only for BiT-128. The higher sets should not be considered unaffected: they instantiate the same invalid joint sampling argument, and

improved nonlinear estimation can use the exact $m(v)$ of (10), the hint, and the hidden-error response correlations.

Breaking the 128-bit parameter set already contradicts the claimed security of the submitted family at that level and supplies a practical UF-CMA forgery.

9 Repair considerations

Suppressing the constant response would remove the easiest sign sensor, but it would not make (4) secret-independent. The remaining response coordinates still share r and retain secret-dependent correlations. Likewise, using more precise scalar acceptance arithmetic or independently resampling mask partitions cannot repair a joint-distribution error.

A repair must change the distributional argument. Possible directions are:

1. use rejection with the true vector mixture

$$\frac{1}{2} \prod_i f_i(z_i - v_i) + \frac{1}{2} \prod_i f_i(z_i + v_i)$$

in the denominator and prove a practical global rejection bound;

2. adopt a standard bimodal rejection construction whose acceptance probability is derived for the complete vector, as in prior work on bimodal lattice signatures [2]; or
3. remove the global bimodal sign and redesign the response and parameter set around a jointly sound unimodal rejection sampler.

Any revised design should test joint moments and latent-variable distinguishers in addition to checking scalar histograms. Matching every one-dimensional marginal is insufficient for zero knowledge or signing-key privacy.

10 Conclusion

BiT’s rejection sampler proves and implements the wrong distributional object. The scalar target is correct, but a sign shared by the full response survives as a latent variable in the joint distribution. The public response to the known constant secret component exposes that variable with high confidence. Posterior-weighted ring regression then recovers s_0 , and public-key compression both corrects the last uncertain coefficients and completes an equivalent secret key.

The resulting attack is practical by a wide margin. The main run required 200,000 signatures and 139 CPU-seconds on an older four-core machine while using one thread. Three of four tested keys forged at that query count, and the fourth forged at 250,000. Security analysis of bimodal rejection must therefore preserve the global coupling throughout; coordinatewise marginal arguments cannot justify a vector signature distribution.

A Independent reproduction procedure

The following procedure is sufficient to reproduce the attack without any private experimental artifact.

1. Obtain the official BiT candidate archive cited in [1], select the BiT-128 reference parameter set, and compile its key generation, signing, signature decoding, challenge expansion, and verification routines. Confirm that public keys and signatures have lengths 1,048 and 1,504 bytes.

2. Generate one key pair. Obtain 200,000 valid signatures on distinct messages. Only the public key and signatures are inputs to the steps below. Decode $(z_{1,0}, z_{1,1}, h, \tilde{c})$ and expand $\tilde{c}$ to the weight-30 polynomial c for each signature.
3. For each of the 30 indices i with $c_i = 1$, read $x = z_{1,0,i}$. Accumulate the log likelihood in (7); if a numerator is zero set $w = -1$, and if a denominator is zero set $w = +1$. Otherwise set $w = \tanh(\Lambda/2)$. As basic checks, decoded x must lie in $[-7, 7]$ and the running mean of w^2 should approach 0.889.
4. Let $\zeta_k = \exp(\pi i(2k + 1)/256)$ and define $\hat{a}_k = \sum_{j=0}^{255} a_j \zeta_k^j$. For each of the three public $z_{1,1}$ polynomials, accumulate $B_{j,k} += w \overline{\hat{c}_k} \hat{z}_{j,k}$. Accumulate the common denominator $D_k += w^2 |\hat{c}_k|^2$.
5. Compute $\tilde{s}_j$ by the inverse negacyclic transform of $B_{j,k}/D_k$ and round every coefficient to the nearest member of $\{-1, 0, 1\}$. No secret-key data is needed to assess this candidate: test whether every coefficient of $A_0 \tilde{s}$ admits a ternary error satisfying (14).
6. If the direct candidate fails, rank coefficients by $||\tilde{s}_{j,i}| - 1/2|$. For the 24 lowest-confidence positions, pair the rounded value with its adjacent ternary value. Enumerate subsets in Gray-code order. Precompute the contribution of each alternative to $A_0 s_0$ and screen each subset on three public-key coefficients before applying the full check.
7. For the passing s_0 , choose e' and calculate b'_0 as in (15). Form a signing key containing the submitted public-key components, s_0, e', b'_0 , a fresh signing-randomness seed, and the hash of the public key. Sign a message that was not among the oracle queries and submit the resulting 1,504-byte signature to the unchanged reference verifier.

The transform accumulators are the only state that grows with the ring degree; the query count affects running time but not attack memory. Repeating the experiment with a different key-generation seed provides an independent key trial without changing any recovery parameter.

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